

Name: _____

7.5/10 marks

Chapter 3 and 6 Test Thinking/Inquiry

Rubric	Level 4+ 10 marks	Level 4 8-9 marks	Level 3 7 marks	Level 2 6 marks	Level 1 5 mark	NOT YET
Thinking/ Inquiry Questions answered using the formulas and units required to demonstrate understanding of the concepts.	All questions answered using the correct formulas and units as required. A thorough and correct understanding of All concepts/ questions demonstrated.	All questions answered using the correct formulas and units as required. A thorough and correct understanding of Most concepts/ questions demonstrated.	All questions answered using most of the correct formulas. A good understanding of most concepts/ questions demonstrated.	Most questions answered or attempted. Questions answered using some of the correct formulas. Some understanding of concepts/ questions demonstrated.	Some questions answered or attempted. Questions answered using limited formulas. Limited understanding of concepts/ questions demonstrated.	Questions not answered or show little understanding of concepts.

Clearly answer the following questions showing all your work. Round your final answer to 3 sd.

1. Determine the frequency and period of rotation of airplane propeller if a spot at the end of a blade is 1.5m from the centre and has a centripetal acceleration of magnitude $2.37 \times 10^3 \text{ m/s}^2$.

$$r = 1.5 \text{ m}$$

$$a_c = 2.37 \times 10^3 \text{ m/s}^2$$

$$a_c = 4\pi^2 r f^2$$

$$f = \sqrt{\frac{a_c}{4\pi^2 r}}$$

$$= \sqrt{\frac{(2.37 \times 10^3 \text{ m/s}^2)}{4\pi^2 (1.5 \text{ m})}}$$

$$= 6.32 \text{ Hz}$$

$$a_c = \frac{4\pi^2 r}{T^2}$$

$$T = \sqrt{\frac{4\pi^2 r}{a_c}}$$

$$= \sqrt{\frac{4\pi^2 (1.5 \text{ m})}{2.37 \times 10^3 \text{ m/s}^2}}$$

$$T = 0.158 \text{ s}$$

inverse of each other

frequency is
6.32 Hz and
the period is
0.158 rev / sec

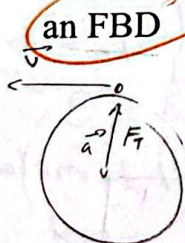
$$T = 0.158 \text{ s}$$

Units

$$T = 0.16 \text{ s}$$

$$f = 6.326 \text{ Hz}$$

2. A 0.25-kg ball attached to a string is rotating in a horizontal circle of radius 0.5 m. If the ball revolves twice every second, what is the magnitude of tension in the string? Include an FBD



$$\sum F_x = F_T = F_c$$

$$F_T = F_c$$

$$m a_c = F_T$$

$$m 4\pi^2 r f^2 = F_T$$

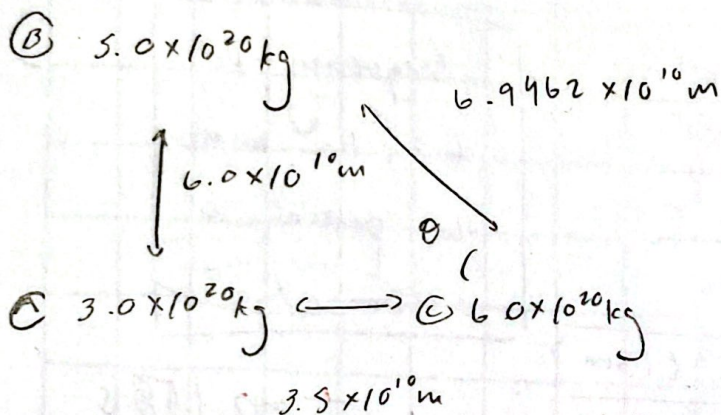
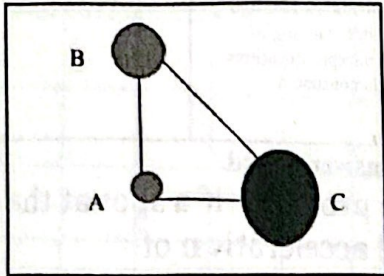
$$F_T = (0.25 \text{ kg}) (\pi^2) (0.5 \text{ m}) (2 \text{ Hz})^2 \times 4$$

$$= 9.9348 \text{ N} \times 4$$

$$19.7 \text{ N}$$

$$19.7 \text{ N}$$

3. The figure shows three large asteroids which are arranged as a right angle triangle. Asteroid A has a mass of 3.0×10^{20} kg. Asteroid B has a mass of 5.0×10^{20} kg and 6.0×10^{10} m away from asteroid A. Asteroid C has a mass of 6.0×10^{20} kg and is 3.5×10^{10} m away from asteroid A. Determine the net F_g on asteroid B from asteroids A and C. Assume distances are from centre to centre.



$$\text{distance } BC = [(6.0 \times 10^{10})^2 + (3.5 \times 10^{10})^2]^{\frac{1}{2}} \quad \theta = \tan^{-1} \left(\frac{6.0 \times 10^{10} \text{ m}}{3.5 \times 10^{10} \text{ m}} \right)$$

$$= 6.9462 \times 10^{10} \text{ m} \quad = 59.74^\circ \left[\leftarrow \uparrow \right]$$

$$F_{gAB} = \frac{(6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2)(3.0 \times 10^{20} \text{ kg})(5.0 \times 10^{20} \text{ kg})}{(6.0 \times 10^{10} \text{ m})^2}$$

$$= 2779166667 \text{ N} \rightarrow y\text{-direction}$$

$$F_{gCB} = \frac{(6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2)(5.0 \times 10^{20} \text{ kg})(6.0 \times 10^{20} \text{ kg})}{(6.9462 \times 10^{10} \text{ m})^2}$$

$$= 9197176523 \text{ N}$$

split into x and y comp.

CONT. ON OTHER PB.

$$F_g = \frac{G m_1 m_2}{r^2}$$

formula used

$$\sum F_x = F_{ABx} + F_{BCx}$$

$$= 0 \text{ N} + (4147176.523 \times \cos 59.7^\circ)$$

$$= 2.092365116 \times 10^9 \text{ N}$$

$$\sum F_y = F_{AB_y} + F_{BC_y}$$

$$= 2779166667 + (4147176.523 \times \sin 59.7^\circ)$$

$$= 6.359820424 \times 10^9 \text{ N}$$

$$\sum F_B = (6.359820424^2 + 2.092365116^2)^{\frac{1}{2}}$$

$$= 6.695170469 \times 10^9 \text{ N} \rightarrow 6.70 \times 10^9 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{6.359820424}{2.092365116} \right)$$

$$= 71.7^\circ \rightarrow 72^\circ$$

W + E
S

$$\text{NET } F_B \text{ on } B = \boxed{6.70 \times 10^9 \text{ N } [W 72^\circ \text{ South}]}$$

East on
right

$$6.69 \times 10^9 \text{ N } [E 72^\circ S]$$

$$\theta = \frac{F_y}{F_x} = \tan^{-1} \left(\frac{6.09}{2.09} \right)$$

$$= 72^\circ$$